

Full length article

Power ramp flash crystallization of a lithium silicate-based glass for dental applications: Structural insights and mechanical properties



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ABSTRACT

This study presents a novel method for the ultrafast crystallization of bulk glasses, termed Power Ramp Flash (PRF) crystallization, which employs a controlled electric field and temperature modulation. We demonstrate the rapid monophasic crystallization of a lithium silicate-based glass, relevant for dental applications. A non-stoichiometric, multicomponent glass-forming composition, similar to Ivoclar e.max®CAD, was melt-quenched into rectangular bar molds and then subjected to crystallization using flash crystallization and a conventional method for comparison. Our results highlight the importance of controlling the power surge during the flash event and the subsequent crystallization process to achieve glass-ceramics with the desired phases and volume fractions, enabled by precise temperature control. Additionally, distinct electrical signatures corresponding to the crystallization of lithium metasilicate and lithium disilicate were identified, demonstrating the potential for *in-situ* studies of crystallization processes. We validate our experimental results using finite element analysis to simulate the sample temperature during flash crystallization, accounting for Joule heating. Comprehensive structural characterization shows that our method crystallizes the disilicate phase within seconds at a furnace temperature of only 360 °C, significantly below the materials' glass transition temperature ($T_{g, onset} = 470$ °C). After optimizing the electrical parameters involved in flash crystallization, specimens were produced featuring mesoscale crystalline domains (~50 nm), a Vickers hardness of approximately 6.80 GPa, and an indentation fracture toughness (K_{IC}) of 1.54 MPa m^{0.5}. Overall, our findings demonstrate, for the first time, the capability of flash crystallization to rapidly crystallize these glass-ceramics in a controlled fashion without compromising their mechanical properties.

1. Introduction

In recent years, the global market for dental prostheses has grown substantially, with an estimated value of USD 7.55 billion by 2024 [1]. This growth has been driven by strong economic expansion in emerging markets, increased healthcare spending, and the rising demand for permanent implants over conventional dentures [1]. As a result, the dental materials community has focused on developing novel biomaterials, particularly for prosthodontic applications, with the primary goals of enhancing the comfort, esthetics, durability, and cost-effectiveness of dental materials while streamlining the prosthetics

workflow [2].

A key motivation for researching new materials is the demand for metal-free solutions in dentistry, as traditional metallic biocompatible materials are often avoided because of their esthetic mismatch with natural teeth and the potential to induce undesirable reactions in the oral environment. Consequently, there is a pressing need for alternative materials that offer improved esthetics, biocompatibility, and overall performance in dental applications [3].

In this context, glass-ceramics, particularly Li₂O–2SiO₂ (lithium disilicate–LS2)-based systems, have gained considerable attention because of their low solubility in the oral environment, excellent

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mechanical properties, high translucency, and esthetics qualities. These attributes make them suitable for various dental applications, including crowns, onlays, inlays, and bridges [4]. Beyond their biocompatibility and mechanical properties, these materials are highly machinable by Computer-Aided Design & Computer-Aided Manufacturing (CAD/CAM) systems [2].

Despite these advantages, processing LS2 glass-ceramics is time-consuming, requiring hours for the nucleation and growth of the metasilicate phase at temperatures of approximately 620 °C, followed by at least 30 min to promote phase transformation to disilicate phases at approximately 830 °C [5]. Additionally, traditional thermal treatments are highly energy-inefficient, as relatively large furnaces must be heated to high temperatures to crystallize relatively small samples. Reducing crystallization time and improving efficiency is crucial for cost-effective material production and essential for further expanding the adoption of LS2 glass-ceramics by the dental industry.

Recently, we introduced an innovative technique capable of performing the sintering and concurrent crystallization of powdered glass compacts in a single step using electric fields [6,7]. This method, termed *flash sinter-crystallization*, involves applying an electric field while heating the glass at a relatively low temperature inside a furnace. The resulting thermal runaway causes the current to rise significantly, leading to rapid volumetric heating, effectively sintering and sometimes crystallizing the material in a very short time. Moreover, the furnace temperatures at which the thermal runaway is triggered are much lower than those used in conventional sintering and/or crystallization treatments (approximately 50 % lower).

In this study, we investigate the possibility of crystallizing bulk glass pieces using, for the first time, a power-controlled version of Flash Crystallization. Our aim is to enhance the process's time and cost efficiency by improving the prosthetics workflow without compromising specimen esthetics, properties, and durability. Optimized crystallization conditions are reported for commercial-like LS2 glass-ceramics, and the flash-crystallized material's microstructural, structural, and mechanical properties are compared with those of conventionally prepared samples.

2. Material & methods

2.1. Synthesis and sample preparation

The specific compositions of commercially available dental glass-ceramics are typically proprietary and not disclosed by the manufacturers. Therefore, the composition used in this study was based on the work of Lubauer et al. [8], who derived the formulations of several commercial LS2 glass-ceramics through chemical analysis using Inductively Coupled Plasma-Optical Emission Spectroscopy (ICP-OES). A total of 80 g of base glass was prepared with the following molar composition, based on the commercial Ivoclar Vivadent IPS e.max®CAD: 68.353 SiO₂ – 24.468 Li₂O – 2.480 K₂O – 1.97 Al₂O₃ – 1.41 P₂O₅ – 0.83 CeO₂ – 0.375 ZrO₂ – 0.055 Na₂O – 0.041 CaO – 0.018 Fe₂O₃.

Reagent-grade powders of SiO₂ (Vitrovita, 99.90 %), Li₂CO₃ (Sigma-Aldrich, 99.00 %), K₂CO₃ (Sigma-Aldrich, 99.50 %), Al₂O₃ (Sigma-Aldrich, 99.90 %), (NH₄)₂HPO₄ (Sigma-Aldrich, 98.00 %), CeO₂ (Sigma-Aldrich, 99.90 %), ZrO₂ (Sigma-Aldrich, 99.90 %), Na₂CO₃ (Sigma-Aldrich, 99.50 %), CaO (Sigma-Aldrich, 99.90 %), and Fe₂O₃ (Sigma-Aldrich, 95.00 %) were used as precursors. The appropriate quantities of these reagents were dried at 120 °C for 24 h and mixed in a polyethylene jar using a SpeedMixer (FlackTek) for 15 min. The resulting batch was then heated at a rate of 10 °C min⁻¹ in an alumina crucible to 700 °C for 2 h to decompose the carbonates (Na₂CO₃, K₂CO₃, Li₂CO₃) and (NH₄)₂HPO₄, releasing NH₃, CO₂, and H₂O. Finally, the calcined mixture was melted at 1400 °C for 30 min (10 °C min⁻¹), and the molten material was cast into a stainless-steel mold, as illustrated in Fig. 1a, and then pressed with a lid, yielding several 45 mm x 5.8 mm x 5.8 mm transparent glass bars with a slight yellow tint (see Fig. 1b). The glass bars were immediately placed in a preheated muffle at 420 °C (50 °C below the glass transition temperature) for 2 h to relieve residual stresses and then cooled to room temperature at a slow rate of 2 °C min⁻¹. Transparent and homogeneous glasses were obtained without visual signs of liquid-liquid phase separation or crystallization.

The glass was then cut into bars (22.5 mm x 5.5 mm x 5.5 mm) and polished on all sides. Fig. 1b depicts the glass specimens at two different

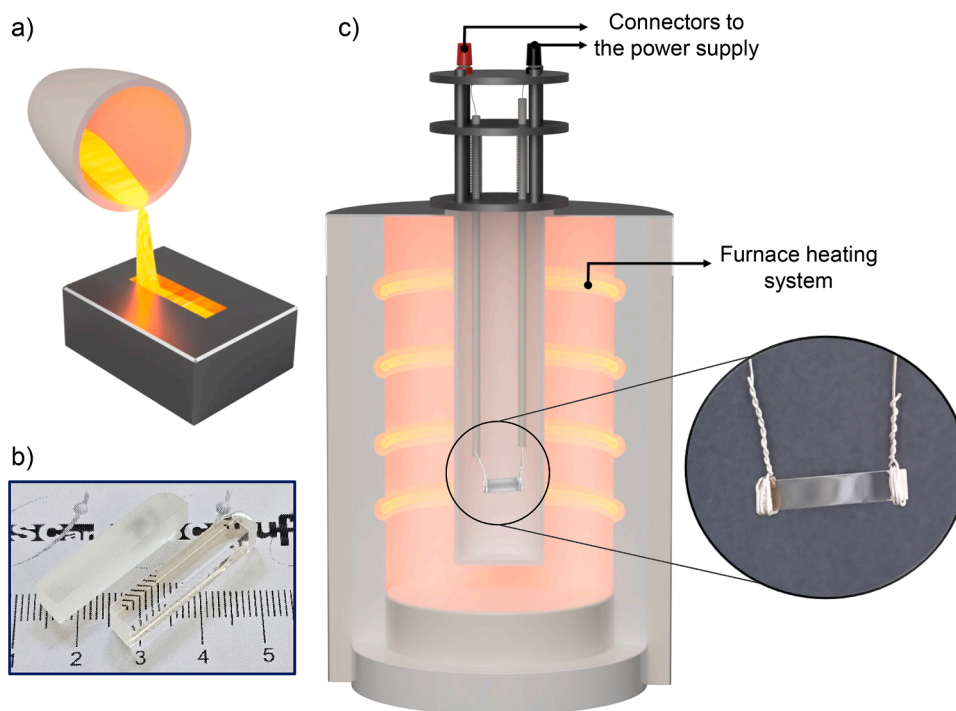


Fig. 1. (a) Illustration of the casting process; (b) specimens at different polishing stages; (c) illustration of the flash crystallization setup, showing the vertical tubular furnace and sample holder. The inset in (c) shows a sample positioned in the sample holder with the silver paste applied to its ends.

stages of polishing.

Conductive silver paste (Sigma-Aldrich, 735,825–25 G) was applied on both ends of the glass bars (as shown in the inset of Fig. 1c) to improve the electrical contact and enable the reversible electrochemical reaction between the cationic conductor (LS2-based glass) and the electrodes (electronic conductors), as previously reported by Caliman et al. [9] and Lachal et al. [10] for other cationic conductors.

2.2. Crystallization process

The glass bars were crystallized through flash crystallization (FC), a process inspired by our previous work [11]. Fig. 1c illustrates the furnace and sample holder designed for FC. The setup consisted of a tubular furnace, a sample holder, a controllable power supply, and a digital multimeter. In all experiments, an AC 1000 Hz (sinusoidal wave) electric field with a magnitude of 180 V cm^{-1} (RMS values) was applied to the specimens, using two silver wires in contact with the silver paste, connected to the controllable power supply (Ametek, Asterion AST 751) and a digital multimeter (Tektronix, Keithley DMM6500). The samples were placed in the furnace pre-heated at 360°C , and the power supply was activated after a 5 min waiting period to ensure uniform temperature distribution within the samples before initiating the flash experiments. A thermocouple was positioned 2 mm away from the specimens to monitor the furnace temperature (T_f) during the FC process.

We investigated three different crystallization protocols: (i) classical [12], (ii) current ramp [13], and (iii) power ramp.

- (i) *Classical Flash (CF)*: The maximum electrical current and subsequent holding time are controlled, while the current rises uncontrollably during the flash event.
- (ii) *Current Ramp Flash (CRF)*: The current during the flash event is increased linearly at a specific rate until its maximum is reached, and it is then held for the designated holding time.
- (iii) *Power Ramp Flash (PRF)*: A power density ramp is employed, with its value regulated every quarter of a second until a preset maximum is reached, after which it is held at that maximum value for a specific holding time.

The first two methods, CF and CRF, are standard approaches in flash experiments [14–18]. They were included in this study to highlight their limitations in controlling temperature rise during crystallization, which can lead to hotspot formation and heterogeneities in the final material. These methods serve as a comparison to emphasize the need for a more refined approach. The PRF method, introduced here for the first time, was specifically developed to enhance the crystallization process by precisely controlling power density increments. This approach ensures a more stable and accurate regulation of sample temperature.

To determine the optimal conditions for the crystallization of the LS2 specimens, we tested a range of maximum current densities from 10 to 40 mA mm^{-2} (in intervals of 10 mA mm^{-2}) and power densities from 10 to 70 mW mm^{-3} (in intervals of 10 mW mm^{-3}), using a current ramp rate of $0.1 \text{ mA mm}^{-2} \text{ s}^{-1}$ for the CF experiments and a power ramp rate of $0.5 \text{ mW mm}^{-3} \text{ s}^{-1}$ for the PRF experiments. After each experiment, we assessed the visual aspects of the samples in terms of transparency and homogeneity. The parent glass and the glass-ceramic with only the metasilicate phase are transparent, while the samples change color upon lithium disilicate crystallization. In all experiments, the maximum current density (CRF and CF) or maximum power density (PRF) was maintained for a holding time of 60 s before the power supply was switched off. The samples were then removed from the furnace and allowed to cool under atmospheric conditions.

Through this optimization, we found that the most homogeneous samples were achieved at a maximum current density of 30 mA mm^{-2} for the CF, 20 mA mm^{-2} for the CRF, and a power density of 50 mW mm^{-3} for the PRF. At lower current or power densities, the samples remained transparent and did not crystallize into the lithium disilicate phase,

while at higher values, the core of the samples began to re-melt. Additionally, we evaluated two different ramp rates for the current, CRF ($1 \text{ mA mm}^{-2} \text{ s}^{-1}$ and $0.1 \text{ mA mm}^{-2} \text{ s}^{-1}$), and the power, PRF ($2 \text{ mW mm}^{-3} \text{ s}^{-1}$ and $0.5 \text{ mW mm}^{-3} \text{ s}^{-1}$). The samples were labeled as CRF1, CRF0.1, PRF2, and PRF0.5, respectively. For comparison, some specimens were conventionally crystallized, i.e., without an electric field, using a single-step thermal treatment at 812°C at a heating rate of $10^\circ\text{C min}^{-1}$ and a dwell time of 1 h (labeled as CC).

In addition to the aforementioned experiments, we conducted a proof-of-concept using our new PRF approach to demonstrate its capability in controlling glass crystallization, producing either the lithium meta- or the di-silicate phase. To this end, a two-step process was designed: initially, the power density was ramped at $0.5 \text{ mW mm}^{-3} \text{ s}^{-1}$ until it reached 25 mW mm^{-3} , followed by a holding time of approximately 180 s to allow the electric field to stabilize. The power density was then increased at the same rate until it reached 50 mW mm^{-3} , where it was held for 100 s.

2.3. Finite element analysis

It is important to note that the sample temperature deviates from the furnace temperature of 360°C during the crystallization experiments because of Joule heating. To estimate the sample temperature during the experiments, we employed a finite element simulation, incorporating power dissipation from Joule heating and heat losses within the furnace. A 2D model was created using COMSOL's physics-controlled module with a "finer-mesh" setting, as illustrated in Fig. 2. The inset highlights the rectangular LS2 glass-ceramic sample (in yellow), with small blue rectangles at the edges indicating the areas coated with silver paste (the electrodes). The surrounding space represents the air within the furnace, and a red dot marks the position of the temperature probe, where the simulated sample temperature ($T_{\text{center, FEA}}$) data were collected.

The material properties used in this model are summarized in Table 1, with values provided at room temperature for simplicity. Temperature-dependent properties were based on data from the indicated references, except for emissivity and the parameters of LS2, which were considered temperature-independent because of a lack of available data. With this approximation, the simulations should be regarded as semi-quantitative, and are used here to qualitatively validate the experimental temperature trends.

Heat losses were accounted for through three mechanisms: conduction, convection, and radiation, using the COMSOL module *Heat sources in solids and fluids*. Conductive heat flux was considered for the materials adjacent to the LS2 sample (and within the sample itself), particularly the silver electrodes. Convective heat flux was modeled between the hot surfaces and surrounding air, considering natural convection. Radiative heat flux was also included based on the emissivity values reported in Table 1.

The heat source in the model was the Joule heating generated by the electric current flowing through the electrodes and the LS2 sample. Experimental power density data, recorded at 0.25 s intervals during the flash experiments, were used as time-dependent inputs for the model. Each experiment's unique power density profile was applied to ensure an accurate representation of the experimental conditions.

The simulation was conducted in the time-dependent domain, starting when the power supply was switched on and continuing until the end of the experiment (typically 2–3 min). The results were recorded at 0.25-s intervals to match the experimental data recording intervals. This approach allowed for an estimation of the temperature within the sample, accounting for the power injected by Joule heating and the varying experimental conditions.

2.4. Characterization

Differential scanning calorimetry (DSC) analysis was performed using a NETSZCH 404 thermal analyzer to characterize the onset glass

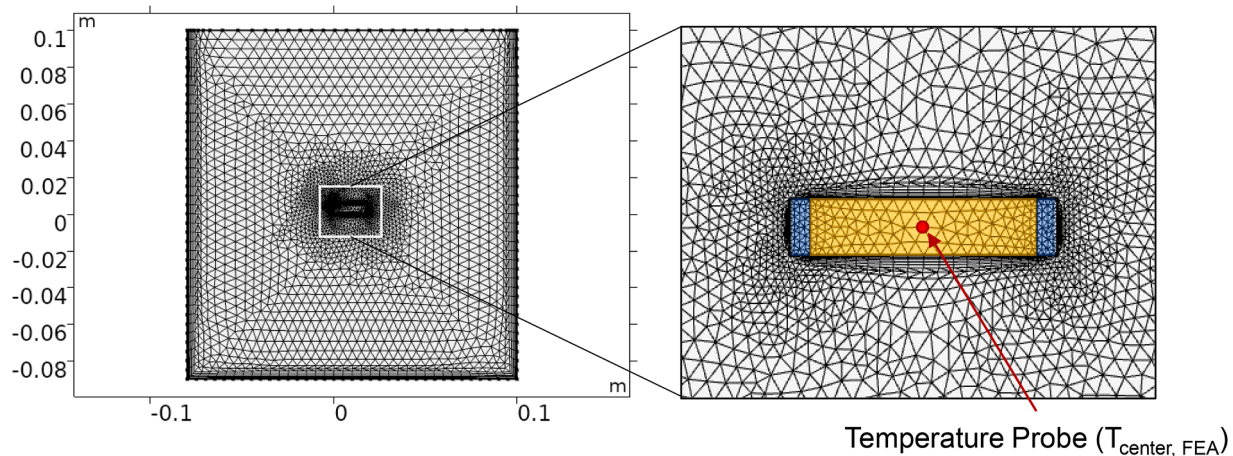


Fig. 2. Mesh details within the furnace and sample during the flash experiments. The inset highlights the mesh of the rectangular sample (in yellow) with silver paste applied at its ends (in blue), along with the position of the temperature probe, where we collected the $T_{\text{center, FEA}}$ data.

Table 1

Properties of the materials used in the Finite Element Model.

Material	Properties	Values at 25 °C	Ref.
Air	Thermal conductivity	0.024 W (m K) ⁻¹	[19]
	Density	0.00123 g cm ⁻³	[20]
	Dynamic viscosity	20.48 μPa s	[21]
Silver	Thermal conductivity	430.0 W (m K) ⁻¹	[22]
	Specific heat capacity	235.5 J (kg K) ⁻¹	[23]
	Density	10.49 g cm ⁻³	[24]
	Emissivity	0.07	[25]
LS2 glass	Thermal conductivity	1 W (m K) ⁻¹	*
	Specific heat capacity	985.62 J (kg K) ⁻¹	[26]
	Density	2.35 g cm ⁻³	**
	Emissivity	0.80	[25]

* Value assumed based on other silicate glasses.

** Calculated using Archimedes' principle.

transition temperature ($T_{g, \text{onset}}$), crystallization peak temperatures (T_c), and liquidus temperature (T_m) of the synthesized glass. A small shard of the parent glass (20 mg) was heated in a platinum pan with a lid, under an N_2 atmosphere at a flow rate of 60 mL min⁻¹, a heating rate of 10 °C min⁻¹, and within a temperature range of 100 °C to 1000 °C.

We employed X-ray powder diffraction (XRD) to analyze the crystalline phases. XRD measurements were conducted on a Rigaku Ultima IV instrument using Cu-Kα radiation, operating at 40 kV and 20 mA, and scanning in the 2θ range of 15–70° with a step size of 0.02° and a scan speed of 2° min⁻¹. Samples were ground in a mortar before the measurements, and the crystal phases were indexed using the ICSD database (Inorganic Crystal Structure Database, V. 4.8.0).

The polished surface of the samples was analyzed using a TESCAN Mira field emission gun scanning electron microscope (FEG-SEM). The glass and glass-ceramics specimens were polished and chemically etched with hydrofluoric acid (10 vol.%) for 30 s. Prior to imaging, a thin layer of gold was sputtered on the polished surface using Quorum Q150R ES equipment.

The electrical conductivity of the specimens was determined by impedance spectroscopy (IS) at different temperatures using a two-electrode setup in air. The surfaces of the specimens were polished and sputtered with gold (Quorum Q150R ES equipment). For the IS measurements, a Novocontrol Alpha-A High-Performance Frequency Analyzer was used, applying a voltage of 300 mV, with the frequency ranging from 10 MHz to 0.1 Hz and the temperature ranging from 250 °C to 100 °C (with five equally spaced measurement points along the temperature axis).

⁷Li satellite transition spectroscopy (SATRAS) [27] was performed at 14.10 T (233.18 MHz) at a magic angle spinning (MAS) frequency of

$\nu_{\text{MAS}}=10.0$ kHz to analyze the spinning sideband patterns arising from the non-central (satellite) ⁷Li Zeeman transitions. The crystalline and vitreous materials of the specimens were investigated using SATRAS difference spectroscopy as previously described [28,29]. Subtraction of a SATRAS spectrum composed of the fast-relaxing vitreous part (obtained by a SATRAS measurement of the sample using a short recycle delay, $d_1 = 1$ s) from the SATRAS spectrum of the fully relaxed sample ($d_1=300$ s) produces a difference spectrum containing only those spectral components not present in the reference, i.e., the ⁷Li ions in crystalline environments. Before subtraction, both spectra were normalized such that the outermost spinning sidebands – arising exclusively from the ⁷Li spins in vitreous environments – matched in intensity. ⁷Li chemical shifts are reported relative to a 1 M LiCl solution.

To investigate the translucency gradient within the specimens, they were cut into 10 slices, each 2 mm thick, using a diamond disk saw. Each slice was sequentially polished with wet silicon carbide paper of increasing grit sizes (240, 400, 600, and 1200), followed by a final polish using velvet fabric soaked in a 0.5 μm cerium oxide suspension. The transmittance of each polished slice was then measured across a wavelength range of 1100 to 190 nm. To quantify the translucency of each slice, the transmittance in the visible range (700 to 380 nm) was integrated.

Vickers microhardness (H_V) and Young's modulus (E_M) were measured at 25 °C and 40 % humidity. The measurements followed ASTM C1327–15 [30] standards using a Micro Combi Tester (Anton Paar GmbH, Germany) and the Oliver and Pharr method [31]. A load of 5 N (with a loading and unloading rate of 10 N min⁻¹) was applied, with a peak holding time of 15 s. Each average microhardness value was calculated from a minimum of 10 indentations on the polished surface of the samples, and E_M was determined by instrumented indentation as a function of the contact depth. The equipment and the indenter tip area function were calibrated before the measurements using a standard fused silica sample provided by the tester manufacturer.

The indentation fracture toughness was calculated according to Eq. (1), proposed by Niihara et al. [32]:

$$K_C = 0.035 \left(\frac{l}{a} \right)^{-0.5} \left(\frac{H_V}{E_M} \right)^{-2/5} \frac{H_V a^{0.5}}{\phi}, \quad (1)$$

where K_C is the indentation fracture toughness (MPa m^{0.5}), l is the crack length (μm) measured from the tip of the indentation until the crack tip, a (μm) is the half-length of the indentation diagonal, H_V (GPa) is the Vickers hardness, E_M (GPa) is Young's modulus, and $\phi = 2.8$ is a dimensionless constant reported by Niihara et al. [32] that can range from 2.7 to 3.

Statistical data analyses were conducted using one-way analysis of

variance (ANOVA) and Tukey's test at a significance level of 5 % ($p < 0.05$).

3. Results & discussion

3.1. Lithium silicate-based glass synthesis

The non-stoichiometric multicomponent glass composition based on the Li_2O – 2SiO_2 system, referred to as LS2, was prepared using the melt-quenching route. The resulting glasses were transparent and homogeneous, showing no visible signs of liquid-liquid phase separation or crystallization, as shown in Fig. 1b, which displays two specimens at different polishing stages after melting and annealing.

Fig. 3 shows the DSC curve obtained from a small piece of the base glass (~20 mg), where four characteristic temperatures were observed. In addition to the onset glass transition, $T_{g, \text{onset}} = 470^\circ\text{C}$, two exothermic peaks were detected at $T_{c,1} = 606^\circ\text{C}$ and $T_{c,2} = 812^\circ\text{C}$, as well as the liquidus temperature at $T_m = 957^\circ\text{C}$. These two exothermic peaks, $T_{c,1}$ and $T_{c,2}$, correspond to the crystallization of lithium meta- and di-silicate phases, respectively. The characteristic temperatures of the LS2 glass reported here are comparable to those of re-melted commercial Ivoclar e.max®CAD metasilicate reported by Lubauer et al. [8]: $T_{g, \text{onset}} = 469^\circ\text{C}$, $T_{c,1} = 642^\circ\text{C}$, $T_{c,2} = 833^\circ\text{C}$, and $T_m = 964^\circ\text{C}$; and by Li et al. [33]: $T_{g, \text{onset}} = 462^\circ\text{C}$, $T_{c,1} = 611^\circ\text{C}$, $T_{c,2} = 790^\circ\text{C}$, and $T_m = 960^\circ\text{C}$. The slight divergence between the references and our results may be attributed to minor differences in the compositions.

Höland et al. [34] reported the influence of P_2O_5 as a nucleating agent on the kinetics of lithium meta- and di-silicate phase crystallization in some commercial LS2 glass-ceramics. They observed that when the P_2O_5 concentration exceeded 3.17 wt.%, the crystallization events of both phases were well separated, whereas they overlapped at lower P_2O_5 concentrations. In our study, the P_2O_5 concentration is 3.54 wt.%, which is consistent with the two distinct crystallization peaks observed in the DSC results.

3.2. Flash crystallization

The LS2 synthesized glasses were subjected to three different types of

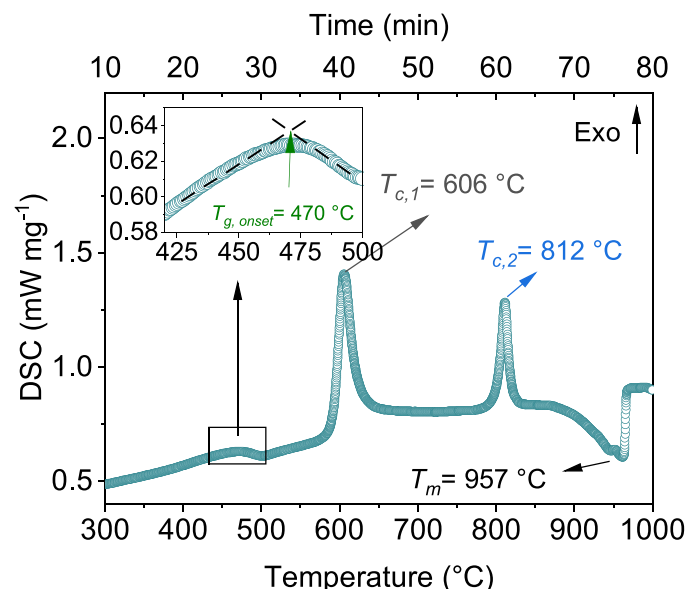


Fig. 3. DSC results of a small piece of the LS2 glass (20.1 mg). The glass transition onset temperature is 470°C , and two exothermic peaks related to the crystallization of the metasilicate phase ($T_{c,1}$) and disilicate phase ($T_{c,2}$) are at 606°C and 812°C , respectively. An endothermic peak at 957°C characterizes the melting onset temperature.

flash experiments: classical flash (CF) [12], current ramp flash (CRF) [13], and power ramp flash (PRF). Fig. 4 illustrates the profiles of electric field (E), current density (J), and power density (P) applied to the LS2 samples during a) CF, b) CRF, and c) PRF. $T_{\text{center, FEA}}$ represents the temperature in the center of the samples, modeled by finite element analysis (see Fig. 2).

For the CF experiments (Fig. 4a), the flash event is easily recognizable by a steep rise in the current density at around 17 s, followed by a sharp drop in the electric field. An overshoot in current density of approximately 15 mA mm^{-2} beyond the set maximum of $J = 30 \text{ mA mm}^{-2}$ is observed, resulting in a power density spike. This spike causes significant power dissipation in the sample in a short period, leading to temperatures exceeding 1400°C (above LS2 melting point, see Fig. 3). This overheating caused hotspots in the CF sample, deforming its original geometry and leading to heterogeneous crystallization, with a glassy phase in the hotspot area and crystallized zones surrounding it.

In the CRF and PRF experiments, current/power is injected at a constant rate (Fig. 4b and c). In these experiments, the electric field initially rises as the sample requires a higher field to maintain the increasing current/power density. Once the current density exceeds $\sim 2.5 \text{ mA mm}^{-2}$, the power dissipated through Joule heating raises the sample temperature sufficiently to significantly reduce its electrical resistivity, triggering thermal runaway and causing a subsequent drop in the electric field. This behavior is sustained as the sample temperature continues to rise until it reaches the crystallization temperature. At this point, a higher electric field is required to maintain the set current/power densities (peaks related to crystallization), suggesting that the sample has become more resistive. In fact, if the ramp rate is sufficiently slow (e.g., Fig. 4b), two distinct peaks in the electric field, corresponding to the crystallization of the meta- and di-silicate phases, can be observed.

Previous studies have also highlighted the relationship between peaks in electrical parameters and changes in material structure during flash experiments in various contexts. For example, time-resolved *in-situ* synchrotron XRD during reactive flash sintering of a multiphase oxide powder revealed that a broad peak in power density corresponded to the phase transition from individual oxides to a single-phase rock salt high-entropy oxide (Mg,Ni,Co,Cu,ZnO) [35]. In another study on the flash synthesis of $\text{CaCu}_3\text{Ti}_4\text{O}_{12}$ from a Pechini-derived amorphous powder, a peak in current density was associated with the crystallization of intermediate phases [36].

Building on the idea that crystallization can be detected during flash experiments through changes in electrical parameters, we designed an experiment to demonstrate the control of crystallization using our PRF approach, as described below.

3.3. Electrical signatures of crystallization events: controlled crystallization

To further investigate whether the peaks observed in the CRF and PRF experiments (Fig. 4b and 4c) are indeed associated with the crystallization of the meta- and di-silicate phases, we conducted a PRF experiment using two power steps. This experiment was designed to separate the crystallization events and demonstrate control over the crystallization process through the PRF method. The power density was ramped at a rate of $0.5 \text{ mW mm}^{-3} \text{ s}^{-1}$, slow enough to distinctly separate the two crystallization peaks. The first step was conducted at a power density of 25 mW mm^{-3} , corresponding to the metasilicate crystallization temperature ($T_{c,1} = 606^\circ\text{C}$). The second step, at 50 mW mm^{-3} , targeted the disilicate crystallization temperature ($T_{c,2} = 812^\circ\text{C}$). After reaching these power densities, the holding time was determined by the stabilization of the electric field, signaling the stabilization of the material's electrical resistivity and, consequently, the crystallization process.

Fig. 5 illustrates the profiles of the electric field (E), current density (J), power density (P), and the estimated temperature ($T_{\text{center, FEA}}$) of the LS2 samples during the two-step approach. The holding times were 180 s

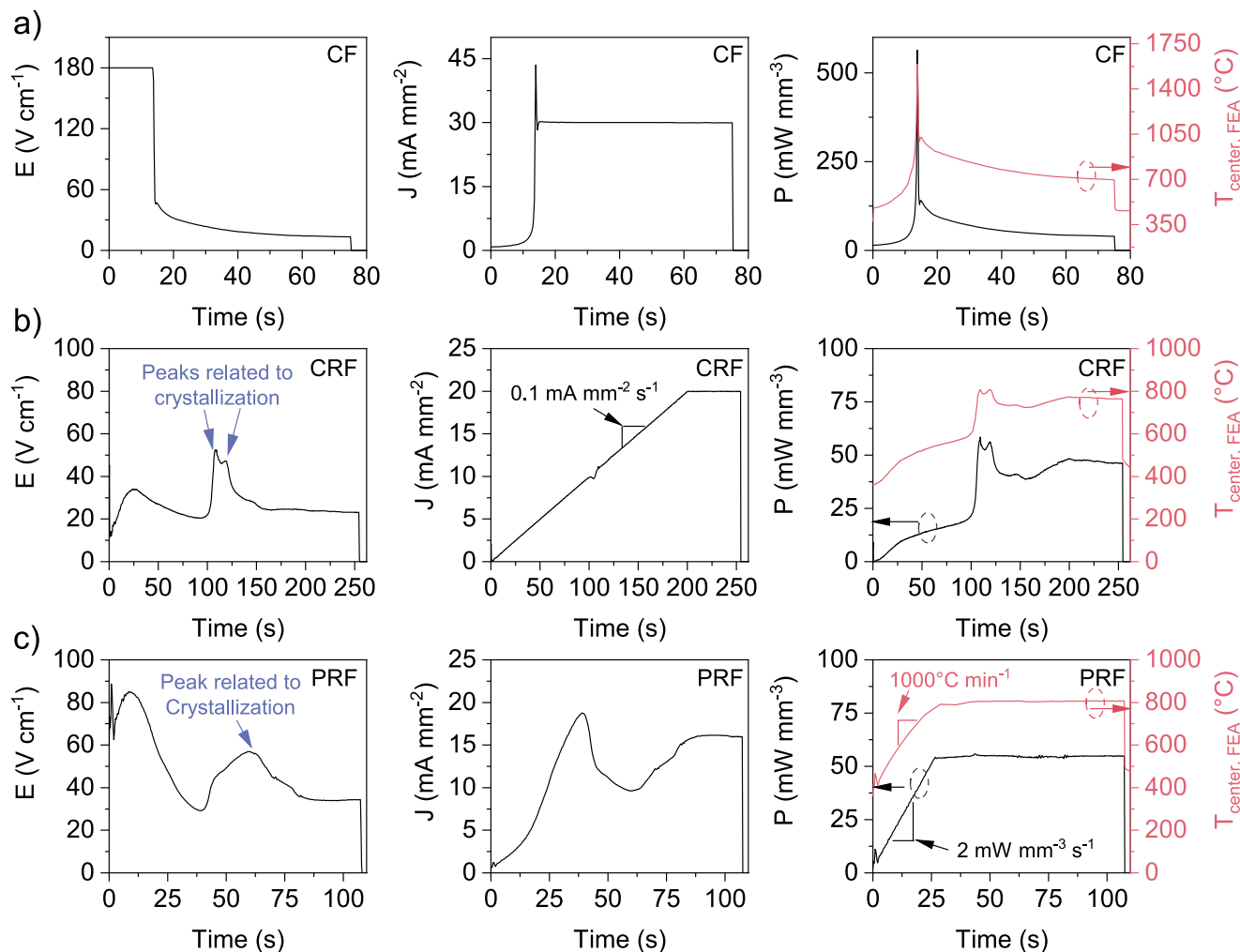


Fig. 4. Flash experiment curves of electric field (E), current density (J), power density (P), and temperature calculated by finite element analysis ($T_{\text{center, FEA}}$) for the (a) classical flash experiment (CF, without ramp control), (b) current ramp flash experiment (CRF), and (c) power ramp flash experiment (PRF).

and 100 s at 25 mW mm^{-3} and 50 mW mm^{-3} , respectively.

In the first power step, the electric field continually rises while the power density remains constant, indicating an ongoing transformation within the material that leads to an increase in resistivity. The electric field then approaches an asymptotic value, signaling that the material transformation has ceased, and the resistivity has stabilized. When the second ramp begins, the electric field drops as the sample temperature increases, consistent with the materials' negative temperature coefficient (NTC) nature (i.e., resistivity decreases as temperature rises). In the second step, the electric field begins to rise again, indicating a second transformation that, like the first, leads to an increase in resistivity. This transformation also stabilizes after some time, as the electric field levels off, indicating that resistivity—and the corresponding material transformation—has reached equilibrium.

To confirm the transformations occurring during both steps, the experiment was stopped at the end of the first step (hereafter referred to as sample (i)) and at the end of the experiment (sample (ii)). *Ex-situ* XRD measurements and impedance spectroscopy (at different temperatures) were then performed, with the results presented in Fig. 6.

Fig. 6a shows the XRD patterns of samples (i) and (ii), alongside the parent glass for comparison. As expected for an amorphous structure, the parent glass exhibits a featureless diffraction pattern, devoid of sharp peaks. In contrast, the XRD data for crystallized samples reveal distinct phases: sample (i) corresponds to the metasilicate phase formed during the first crystallization step, while sample (ii) displays the disilicate phase developed in the second step. The XRD peak widths for the

metasilicate phase are significantly broader than those for the disilicate phase, indicating a smaller crystallite size for the former. Additionally, the crystalline volume fraction is likely different, given that the amorphous band is much more prominent in the pattern of sample (i). These two characteristics explain the higher translucence of sample (i) compared to sample (ii) (see inset in Fig. 5).

From the *ex-situ* impedance spectroscopy measurements of samples (i) and (ii), we extracted the total conductivity values at different temperatures. For completeness, the IS curves of two distinct formalisms (complex impedance and modulus) are presented in the Supplementary Material (Figs. S1, S2, and S3) for the different samples and temperatures. Fig. 6b shows the Arrhenius plot of conductivity for samples (i) and (ii), along with the parent glass for comparison. Additionally, we included the *in-situ* conductivity extracted from the flash experiment data, i.e., the conductivity calculated using (J and E) collected during the two-step PRF, accounting for the phase shift between voltage and current in the AC signal. In this case, the temperature used to build the Arrhenius plot was the estimated sample temperature, $T_{\text{center, FEA}}$.

Sample (i), after metasilicate formation, exhibits a conductivity value lower than that of the parent glass but higher than that of sample (ii) (after disilicate formation). Therefore, during the first step of PRF, while the sample crystallizes into the metasilicate phase, it requires a higher electric field to maintain the constant power density, consistent with the results in Fig. 5. Similarly, in the second step, in which the sample crystallizes into the more resistive phase (disilicate), a higher electric field is required to maintain the constant power (see Fig. 5).

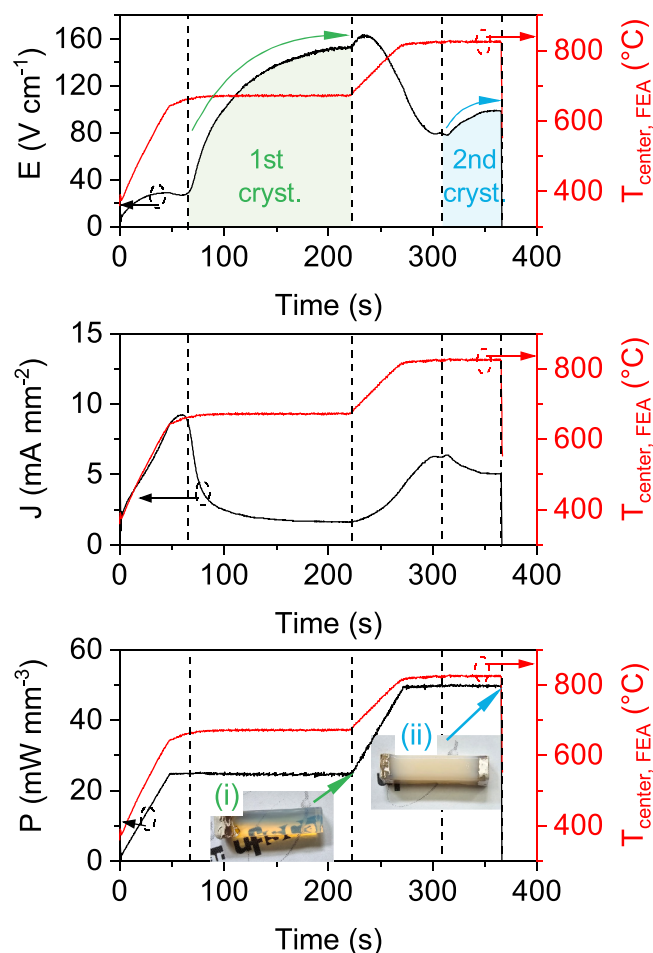


Fig. 5. Flash curves of the electric field (E), current density (J), power density (P), and sample temperature estimated by FEA ($T_{\text{center, FEA}}$) for the sample subjected to PRF using two steps of maximum power density. The insets in the power density curve show two pictures of the sample at the end of each step, labeled (i) and (ii). The dashed lines indicate the beginning and ending of the crystallization zones.

Furthermore, as seen in Fig. 6b, the conductivity calculated using data collected during the PRF experiment (*in-situ*) exhibited significant changes at two points corresponding to the two power density steps (near $T_{c,1}$ and $T_{c,2}$). By comparing the *in-situ* results with the extrapolation of the *ex-situ* IS experiments (dashed lines in Fig. 6b), a good match was observed for the conductivity values from the metasilicate (above $T_{c,1}$) and disilicate (above $T_{c,2}$) phases. It is important to note that the *in-situ* measurements were not performed in the linear range of electrical excitation, i.e., a high electric field was applied [37]. Additionally, during flash events, ionic conductors experience an increase in electronic conductivity because of a reduction in the oxidation state of certain elements, caused by the relatively high electric field being applied [38–42]. Therefore, the extrapolation of *ex-situ* measurements could not account for this change in the conduction mechanisms. This effect appears to be particularly significant for temperatures below $T_{c,1}$, as a considerable difference is detected between the conductivity of the parent glass measured *in-situ* and the values extrapolated from the *ex-situ* data.

In summary, with this two-step PRF approach, we demonstrated that it is possible to control crystallization by monitoring changes in the electrical parameters. To separate the crystallization of the meta- and disilicate phases, we first selected a power step corresponding to the crystallization temperature of the metasilicate. The stabilization of the electric field indicated that sample conductivity and, consequently,

crystallization had reached equilibrium, resulting in a single-phase metasilicate sample. Next, the power was increased to induce the crystallization of the disilicate phase, with the process stabilization again indicated by the electric field stabilization. The correlation between electrical parameters and crystallization opens new possibilities for studying rapid crystallization *in-situ* and could be further explored for other glass-ceramic systems.

Furthermore, for glasses with well-characterized crystallization kinetics, the two-step PRF approach offers a versatile method for double-heat treatments, enabling precise microstructural control of the final glass-ceramic. By carefully adjusting the holding time and power density during each PRF step, it is possible to regulate the nucleation and growth of crystals, thereby achieving a tailored microstructure and desired crystalline volume fraction. Moreover, an n-step PRF approach can be easily adapted for glasses with n-crystallization phases.

Having established the effectiveness of the PRF method in controlling crystallization and the underlying mechanisms, we now focus on characterizing the glass-ceramics produced through PRF at different ramp rates and comparing them to those produced by CF, CRF, and the conventional method.

3.4. Homogeneity of the samples and its relation to hotspots

Fig. 7 shows the parent glass alongside the samples after crystallization using the three flash approaches (CF, CRF, and PRF) and conventional crystallization without an electric field (CC). The samples prepared by CF and CRF exhibited deformation caused by hotspots during crystallization. In contrast, the samples prepared using PRF retained the original geometry of the parent glass, similar to those crystallized via CC.

During the flash experiments, the homogeneous glass initially heats uniformly as the current flows evenly. However, crystallization can begin locally at nucleating sites, causing a sharp rise in resistivity in those regions. As a result, the current will preferably flow through the remaining glassy phase, where the conductivity is higher. In both CF and CRF, the current continues to increase, concentrating heat in the glassy regions, which leads to hotspots and deformation. Although a slower current ramp in CRF (as seen in CRF1 vs. CRF0.1 in Fig. 7) could reduce localized overheating and deformation, only the PRF method effectively prevents hotspots. Since PRF regulates power rather than current, as the conductivity decreases during crystallization, the current automatically adjusts to maintain a constant power ramp, thereby avoiding localized overheating. This self-regulating mechanism in PRF minimizes deformation and ensures more homogeneous crystallization throughout the sample.

3.5. Structural and optical characterization

The X-ray diffractograms of the parent glass, CC sample, and PRF samples with 2 and 0.5 $\text{mW mm}^{-3} \text{ s}^{-1}$ (PRF2 and PRF0.5, respectively) are shown in Fig. 8. All the glass-ceramics feature $\text{Li}_2\text{Si}_2\text{O}_5$ as the main phase. The CC sample exhibits the secondary phases of α -quartz (SiO_2) and Li_3PO_4 . A minor peak related to Li_3PO_4 is also observed in the PRF samples, being more evident for PRF0.5. For comparison, the XRD patterns of the CRF and CF samples are included in the Supplementary Material (Fig. S4). These samples show an amorphous halo beneath the crystalline peaks of the metasilicate (minor) and disilicate (main) phases, along with secondary phases of Li_3PO_4 and $\text{LiAlSi}_2\text{O}_6$. The presence of a significant residual glass phase, metasilicate, and the formation of $\text{LiAlSi}_2\text{O}_6$ in these samples likely result from inhomogeneous temperature distribution, i.e., the formation of hotspots, which is consistent with the observations in Fig. 7. Because of the inhomogeneity observed in the CF and CRF samples, these were not subjected to further characterization. These results highlight the necessity of the refined PRF method for achieving homogeneous bulk crystallization during flash experiments.

Quantifying the crystalline volume fraction of glass-ceramics is

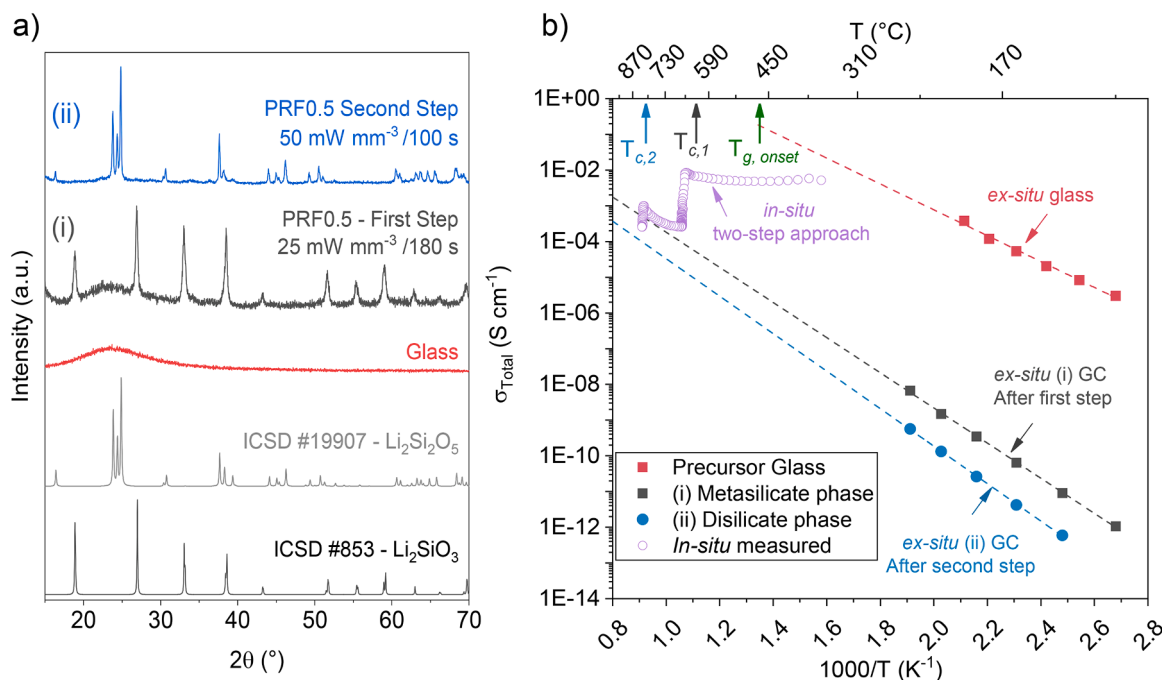


Fig. 6. The two-step PRF approach results: a) ex-situ XRD diffractograms of the parent glass and the two glass-ceramics at stages (i) and (ii), with ICSD reference patterns for the metasilicate and disilicate phases; and b) Arrhenius plot for the total ionic conductivity measured for the parent glass and the (i) and (ii) glass-ceramics by IS, and calculated using the electric field and current density flash data for the in-situ experiments. The dashed line in the Arrhenius plot shows the extrapolation of the data for comparison with the in-situ data at high temperatures.

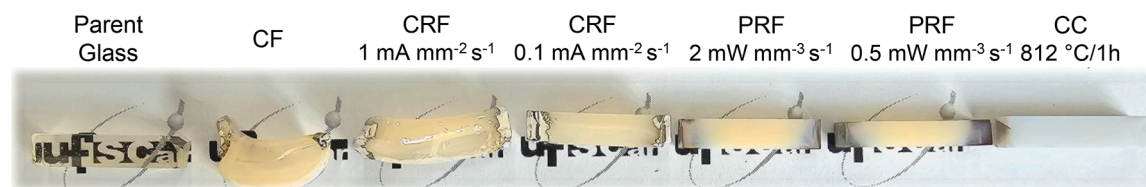


Fig. 7. Images of the parent glass and the specimens after crystallization, with and without an electric field, using different methods and ramp rates.

challenging, particularly for non-stoichiometric compositions and specimens with high crystalline content [43]. Typically, Rietveld analysis of XRD data is used [5], providing an accuracy of about 3–5 % [44]. However, this method requires internal standards, well-known structures as a starting point for refinement, and careful sample preparation to avoid preferential orientation of the crystals. Moreover, it is not well suited for samples containing nanocrystalline or non-stoichiometric phases. Solid-state NMR spectroscopy, on the other hand, can quantitatively detect (nano)crystalline phases in glass-ceramics [45], but only for compositions containing NMR-active isotopes. For example, in silicate glasses and glass-ceramics, the ^{29}Si isotope has a low natural abundance ($\sim 4.7\%$) and long spin-lattice relaxation times, which often make quantitative experiments unfeasible.

In this study, we employed ^7Li NMR spectroscopy – previously shown to sensitively detect subtle structural changes during relaxation, nucleation, and crystallization of lithium disilicate glasses [28]. Furthermore, this method was demonstrated to quantify the crystalline volume fraction of non-stoichiometric Li-based glass systems [29]. Here, we employed ^7Li SATRAS difference spectroscopy to estimate the crystalline fraction of all Li-containing phases: subtraction of the SATRAS spectrum of the base glass from that of the glass-ceramic yields a difference spectrum composed exclusively of the ^7Li nuclei in crystalline environments. Comparing the difference intensity to the whole glass-ceramic spectrum enables quantitative determination of the fraction of Li atoms in crystalline environments. It should be noted, however, that the experimental conditions must be carefully controlled to

obtain quantitative SATRAS spectra, and the method cannot readily distinguish different Li-containing crystalline phases, such as $\text{Li}_2\text{Si}_2\text{O}_5$ and Li_3PO_4 in this study.

Fig. 9 shows the SATRAS (left) and SATRAS difference (right) spectra for the base glass and selected glass-ceramics. The base glass spectrum presents a set of spinning sidebands spanning a frequency range of ~ 250 kHz, approximately 100 kHz narrower than previously observed for stoichiometric LS2 glass [28]. Crystallization of the glass results in significantly narrower spinning sideband manifolds corresponding to ^7Li in crystalline environments. The difference spectra (Fig. 9, right) of all glass-ceramic specimens show that the fraction of lithium ions in the crystalline phase (β) is comparable across the different PRF samples (PRF0.5 and PRF2) and the CC sample, with values of $87.6 \pm 2.0\%$, 91.4

$\pm 2.0\%$, and $90.5 \pm 2.0\%$, respectively. Based on the β -values, the crystalline volume fraction can be derived by considering the composition and density of the parent glass, crystalline phase, and residual glass. For simplicity, we will neglect the minor amount of Li_3PO_4 detected in XRD (see Fig. 8) and only consider $\text{Li}_2\text{Si}_2\text{O}_5$ in the analysis. The volume of the glass-ceramics (V_{GC}) and the volume of $\text{Li}_2\text{Si}_2\text{O}_5$ (V_{LS2}), can then be calculated using the mass of the glass-ceramic (m_{GC}) and the mass of the crystallized phase (m_{LS2})—determined by the molar mass of the components and the fraction of Li_2O (β) and SiO_2 (implicitly) converted to $\text{Li}_2\text{Si}_2\text{O}_5$ (M_{LS2})—, the measured glass-ceramic density (ρ_{GC}) (see Table 1), and the crystallized phase density ($\rho_{\text{LS2}} = 2.52 \text{ g cm}^{-3}$).

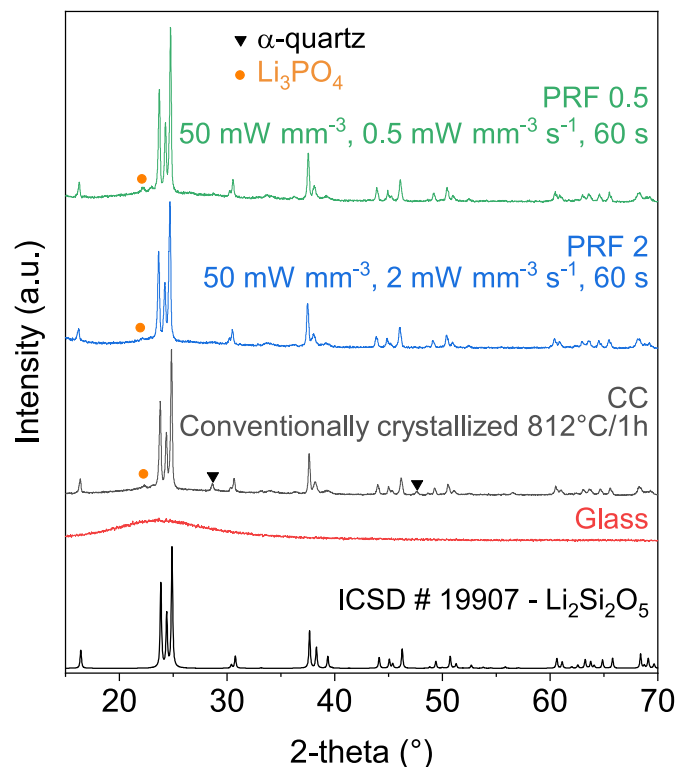


Fig. 8. XRD patterns of the LS2 parent glass and glass-ceramics prepared conventionally (CC), and by flash crystallization using PRF with ramp rates of 0.5 and 2 mW mm⁻³ s⁻¹ (PRF 0.5 and PRF 2, respectively). The bottom diffraction pattern corresponds to Li₂Si₂O₅ (ICSD 19,907) for comparison.

Finally, the crystalline volume fraction of Li-containing crystal phases (α) of the glass-ceramics can be calculated by the following equation (Eq. (2)):

$$\alpha = \frac{V_{LS2}}{V_{GC}} = \frac{\frac{m_{LS2}}{\rho_{LS2}}}{\frac{m_{GC}}{\rho_{GC}}} = \frac{m_{LS2} \rho_{GC}}{\rho_{LS2} m_{GC}}, \text{ with } m_{LS2} = [\text{Li}_2\text{O}] \beta M_{LS2}. \quad (2)$$

Applying Eq. (2) to the β -values obtained from SATRAS, the LS2 crystalline volume fraction of the flash-crystallized specimens PRF2 and PRF0.5 are 57.5 ± 1.2 %, and 55.1 ± 1.2 %, respectively, compared to 57.1 ± 1.2 % for the CC specimen, which means a volume fraction of residual glass phase of 42.5 % for PRF2, 44.9 % for PRF0.5, and 42.9 % for the CC. Therefore, no significant difference in LS2 crystalline volume fraction was observed between the CC and PRF2 samples, while the PRF0.5 specimen exhibited a slightly smaller volume fraction. Moreover, the calculated crystalline volume fraction of LS2 is comparable to that obtained by Lubauer et al. [8] using XRD, which was approximately 60 % for the studied composition (IPS e.max® CAD).

Another interesting feature of the flash-crystallized specimens is the thermal gradient generated by volumetric heating of the sample in a cooler environment, i.e., when the furnace temperature is lower than the sample temperature. Fig. 10a and b illustrate the temperature profile of the PRF2 at different stages of flash crystallization, modeled using finite element analysis (FEA), and highlight the temperature gradient (see also Video S1 in the Supplementary Material).

As depicted in Fig. 10, the extremities of the glass-ceramic bars prepared by PRF are subjected to lower temperatures than the core. To assess the impact of this thermal gradient on transmittance, the sample was sliced transversely into 10 pieces, each 2 mm thick, and UV-VIS analysis was performed on each slice to quantify the transmittance at different points. Fig. 11a and b show the UV-VIS spectra of each slice and the integral of transmittance across the visible wavelength range, respectively. The integral of transmittance was used to facilitate the visualization of the transmittance gradient within the PRF2 sample.

Comparing Fig. 11b with Fig. 10b, it becomes evident that the transmittance gradient is inversely related to the temperature. In other words, regions that experienced lower temperatures during the PRF process exhibit greater translucency. This can be explained by the fact that at lower temperatures, the crystallized volume fraction and crystal sizes are likely smaller, resulting in a more translucent region. Previous studies addressing flash sintering have shown that this temperature gradient can be avoided by thermally insulating the material [46–48].

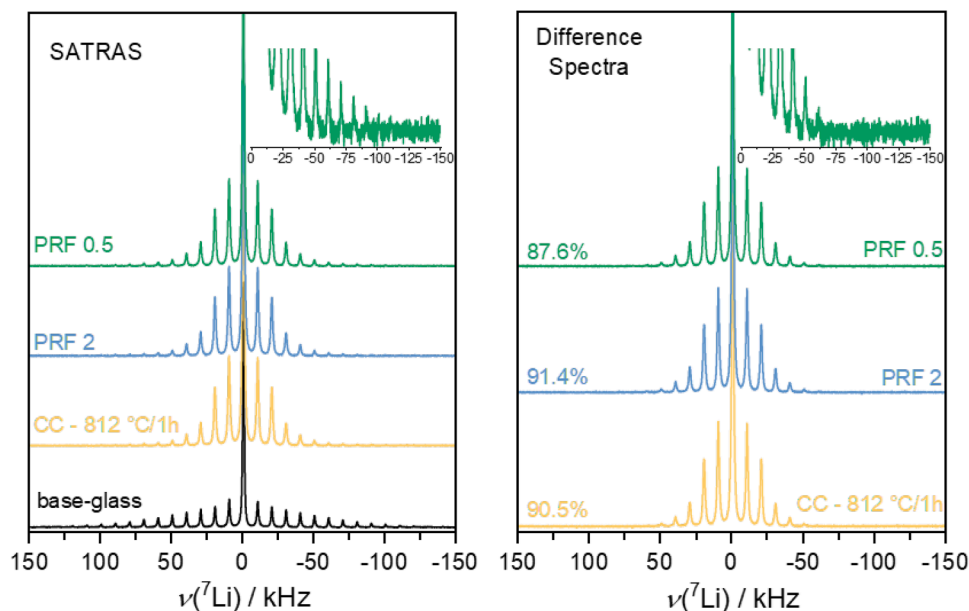


Fig. 9. ⁷Li SATRAS NMR spectra (left) and SATRAS difference spectra (right) for selected specimens produced under the conditions depicted in the plot. The insets are vertical expansions, revealing the absence of the spinning sidebands of the vitreous material within noise levels (approximately -75 to -100 ppm range) across the different spectra. The percentage values indicate the remaining signal area after subtracting the reference (base glass), corresponding to ⁷Li in crystalline environments.

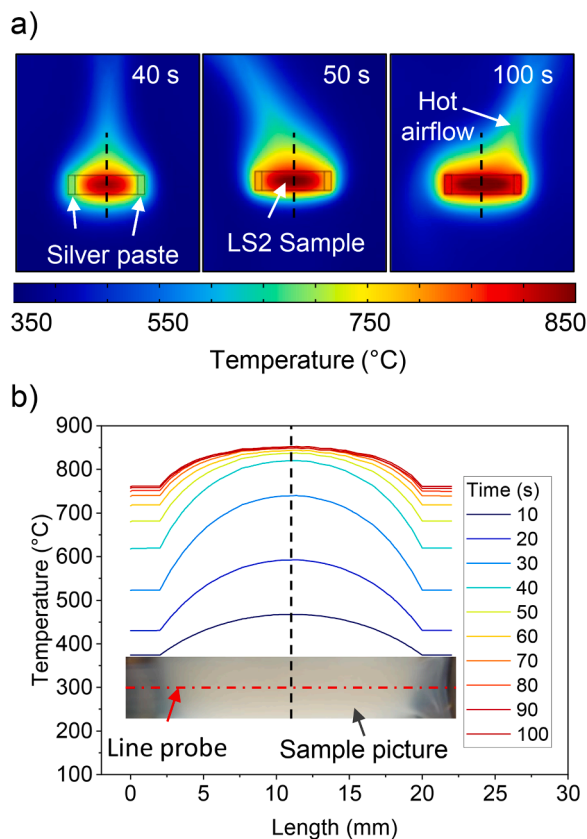


Fig. 10. Finite element analysis of LS2 glass-ceramic produced by PRF with a ramp rate of $2 \text{ mW mm}^{-3} \text{ s}^{-1}$: a) three different frames of the thermal profile modeled by FEA at 40, 50, and 100 s; b) line probe temperature curve in the middle of the sample (red dash-dotted line) acquired at different moments of PRF, highlighting the thermal gradient formed within the sample. The dashed lines mark the center of the samples.

However, this gradient could be beneficial for producing translucency gradients, which are highly desirable for dental prosthesis applications, as they mimic the esthetics of natural teeth.

3.6. Microstructure and mechanical properties

The microstructures of the PRF0.5, PRF2, and CC specimens were examined using SEM (secondary electron detector) on polished and chemically etched surfaces. Fig. 12 presents the microstructure of the selected specimens at various magnifications. The surfaces were etched for 30 s in hydrofluoric acid, revealing a high fraction of spherical nanosized crystals and a few fine, elongated crystal clusters with seemingly random orientations. No significant differences in the microstructures were observed across the different crystallization methods.

Typically, the microstructure of lithium disilicate glass-ceramics is dominated by elongated, acicular crystals that are randomly dispersed and can reach several micrometers in length, depending on the heat treatment time and temperature [8]. However, the formation of spherical crystals in multicomponent lithium disilicate glass-ceramics is not uncommon [49]. Sun et al. [49] investigated the effects of different heat treatments and holding times on the crystallization of multicomponent lithium disilicate glass-ceramics and found that spherical crystals tend to form at temperatures below 850°C , while elongated, rod-like structures develop at higher temperatures. Mavriqi et al. [50] reported a microstructure comparable to that observed in Fig. 12 when crystallizing Ivoclar IPS e.max® at 840°C .

Table 2 summarizes the results of Young's modulus (E_M), Vicker's microhardness (H_V), and indentation fracture toughness (K_{IC}) of the

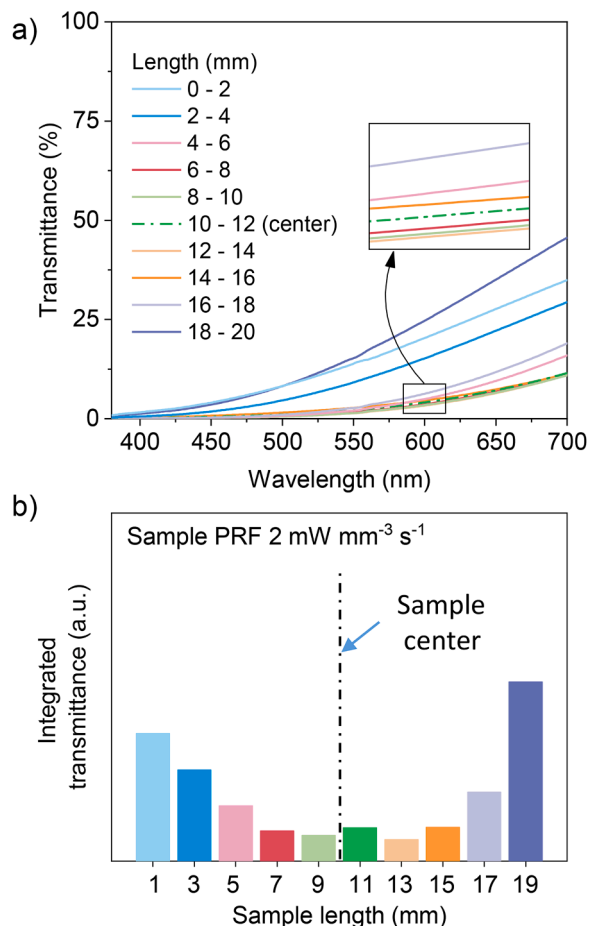


Fig. 11. UV-VIS spectra of each slice of LS2 glass-ceramic produced by PRF with a ramp rate of $2 \text{ mW mm}^{-3} \text{ s}^{-1}$: a) transmittance curves vs. the light wavelength; b) plot of the integral of transmittance for better visualization of the translucency gradient. The inset shows a picture of the LS2 sample after crystallization using PRF2, highlighting the correlation between the translucency gradient and the thermal gradient. The dash-dotted lines mark the center of the samples.

selected glass-ceramics obtained by PRF0.5, PRF2, and CC. As expected from the similar microstructures and crystalline volume fractions, the mechanical properties of the samples flash-crystallized by PRF2 and PRF0.5, as well as those conventionally crystallized, are quite similar, with no statistical differences observed for indentation fracture toughness and Vicker's microhardness (see Table 2). All glass-ceramics exhibited higher Young's modulus, microhardness, and fracture toughness compared to the parent glass, and performed similarly to commercial glass-ceramics in terms of Young's modulus ($\sim 103 \text{ GPa}$ [8]) and microhardness ($\sim 5.3\text{--}6.9 \text{ GPa}$ [51,52]). However, Young's modulus of PRF0.5 was found to be statistically different from those of PRF2 and CC, likely due to the lower crystallized volume fraction in PRF0.5 compared to the other two samples.

The indentation fracture toughness (K_{IC}) for commercially available lithium-based glass-ceramics ranges from 1.45 to $2.3 \text{ MPa m}^{0.5}$ [8], and is related to crystal size, morphology, and crystalline volume fraction. Larger, elongated crystals in an interlocked microstructure typically resist crack propagation more effectively. The fracture toughness values measured here ($\sim 1.5 \text{ MPa m}^{0.5}$) are comparable to those of several commercial dental glass-ceramics, such as Celtra® Duo, Suprinity® PC (crystallized), Initial™ LiSi Block, and N!CE®, as measured using the ball-on-three-balls (B3B) method by Lubauer et al. [8]. However, to increase the fracture toughness to the level of IPS e.max® CAD ($\sim 2.0 \text{ MPa m}^{0.5}$), larger crystal sizes would be required. For future studies, we

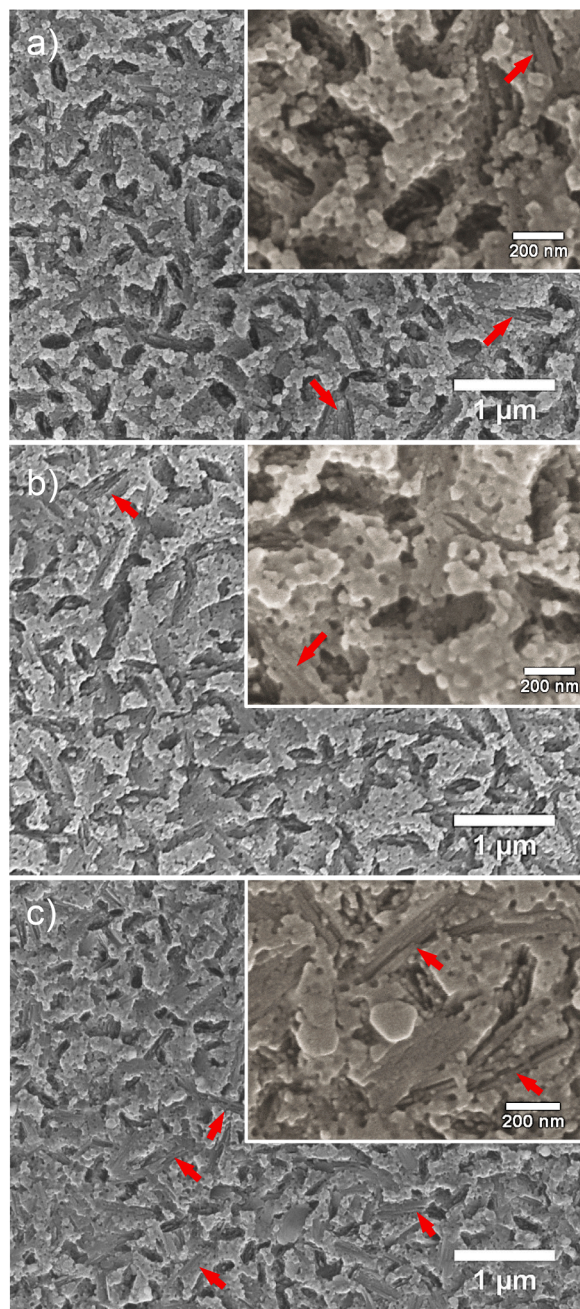


Fig. 12. SEM images of LS2 glass-ceramics at different magnifications after crystallization using (a) PRF2, (b) PRF0.5, and (c) conventional crystallization at 812 °C for 1 h. The red arrows indicate elongated crystal clusters.

Table 2

Mechanical properties of Young's modulus (E_M), Vicker's Microhardness (H_V), and indentation fracture toughness (K_{IC}) of selected glass-ceramics and the parent glass. Values in parenthesis are the standard deviation. Groups sharing the same letter show no statistically significant difference according to Tukey's test at a 5 % significance level.

Glass-ceramic	E_M (GPa)	H_V (GPa)	K_{IC} (MPa m ^{0.5})
PRF0.5	97(1) ^a	6.8(0.1) ^a	1.51(0.03) ^a
PRF2	107(2) ^b	6.8(0.1) ^a	1.54(0.04) ^a
CC	109(2) ^b	6.8(0.2) ^a	1.54(0.04) ^a
Glass	79(3) ^c	5.3(0.2) ^b	1.08(0.04) ^b

suggest exploring different heating strategies using PRF to promote larger crystal growth in LS2, possibly through a double-stage treatment at temperatures where growth is favored, i.e., above 850 °C [49,50].

Finally, an interesting observation is that all specimens produced via flash crystallization exhibited a yellow tint, whereas the conventionally crystallized specimen was predominantly white. This yellowish coloration in the flash-crystallized samples does not appear to be related to the electrode material, as no traces of silver were detected during the SEM with live EDS mode (detection limit of 0.1 wt.%). The EDS results for the CC, PRF1, and PRF0.5 samples are provided in the Supplementary Material (Fig. S5). Since no significant differences in crystalline phases or microstructure were observed between these PRF and CC glass-ceramics, we speculate that the color variation may be related to changes in the oxidation states of transition metal ions. These changes are likely facilitated by the electric field and the rapid heating and cooling cycles during flash crystallization. To investigate this hypothesis, we performed heat treatments on yellowish samples crystallized by PRF2 at 812 °C, in an inert atmosphere (N₂) and air atmosphere. The sample heat-treated in air turned white, while the one treated in the inert atmosphere retained its yellowish color. This outcome suggests that the color variation is likely linked to differences in oxidation states (see Fig. S6 in the Supplementary Material). Previous studies have shown that flash sintering can increase the concentration of oxygen vacancies in polycrystalline materials, leading to cation reduction [53–56]. However, it remains unclear whether the generation of oxygen vacancies is primarily driven by rapid thermal cycling or by the electric field and current passing through the material during flash processing. Further investigation is needed in future studies.

4. Conclusion

We developed a new method, dubbed power ramp flash (PRF), for ultrafast and controlled crystallization of lithium silicate (LS2)-based glass-ceramics. This novel technique enables precise control over the crystallization process by regulating power density, ensuring uniform crystallization throughout the sample, and avoiding issues such as hot-spots commonly encountered in traditional flash methods. Beyond its practical benefits, PRF demonstrated that in the investigated glass, different crystallization steps produce distinct electrical conductivity signatures, offering valuable insights into phase transformations. The glass-ceramics produced via PRF exhibited mechanical properties comparable to conventionally crystallized (CC) samples, with Vickers microhardness values (~6.8 GPa) and indentation fracture toughness (~1.54 MPa m^{0.5}) within the range of commercially available lithium disilicate dental materials. These properties are attributed to a microstructure dominated by spherical nanosized crystals and fine-elongated clusters. The method also shows promise for producing esthetically appealing glass-ceramics with a natural, tooth-like color and translucency gradient, making it particularly relevant for dental applications. By controlling crystallization kinetics, PRF presents a versatile, rapid, and energy-efficient synthesis route for advanced glass-ceramics.

CRediT authorship contribution statement

João V. Campos: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Isabela R. Lavagnini:** Writing – review & editing, Methodology, Investigation, Formal analysis. **Thalia D. de Souza:** Writing – review & editing, Validation, Methodology, Formal analysis, Data curation. **Láís D. Silva:** Writing – review & editing, Formal analysis, Data curation. **Lílian M. Jesus:** Writing – review & editing, Visualization, Supervision, Methodology, Formal analysis, Conceptualization. **Henrik Bradtmüller:** Writing – review & editing, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Edgar D. Zanotto:** Writing – review & editing, Visualization, Supervision, Resources, Conceptualization. **Ana**

C.M. Rodrigues: Writing – review & editing, Visualization, Supervision, Resources, Project administration, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.actamat.2024.120620](https://doi.org/10.1016/j.actamat.2024.120620).

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